

DIVISIBILITY RULES & PROBLEM TYPES

Divisibility Rules

Rule 1: 1 is not divisible by any number except 1. But 1 is a universal factor.

Rule 2: Last digit should be even or divisible by 2.
Ex: 1658 (Even), 24632 (Even), 4359 (Not even)

Rule 3 & 9: Sum of digits should be divisible by 3 and 9.
Ex: $342 \rightarrow 3+4+2 = 9$. $9/3 \rightarrow 3$

Rule 4 & 8: Last 2 digits divisible by 4, and last 3 digits divisible by 8.
Ex: $1624 \rightarrow 24/4$. $23408 \rightarrow 08/8$.

Rule 5 & 10: Any number ends with '5' is div by 5 (Ex: 105). If last digit ends with '0', it is div by 10 (Ex: 1050).

Rule 6: Number should be divided by both 2 and 3.
Ex: $126 \rightarrow \text{even (2)}. 1+2+6=9/3 \text{ (3)}. \text{Then } 126 \text{ is divisible by } 6$.

Composite Rules:

Rule 12 $\rightarrow$ Div by both 3, 4

Rule 14 $\rightarrow$ Div by both 7, 2

Rule 15 $\rightarrow$ Div by both 5, 3

Rule 18 $\rightarrow$ Div by both 9, 2

Rule 7, 13, 17, 19 Osculator methods:

7: $343 \rightarrow 34 - (3 \times 2) = 34 - 6 = 28$. (Div by 7). OR $380247 \rightarrow 380 - 247 = 133$.

13: $385 \rightarrow 38 + (5 \times 4) = 38 + 20 = 58$. (Div by 13). Then 385 is div by 13.

17: $1904 \rightarrow 190 - (4 \times 5) = 190 - 20 = 170$. (Div by 17).

19: $361 \rightarrow 36 + (1 \times 2) = 36 + 2 = 38$. (Div by 19).

Rule 11: (Sum of odd places) - (Sum of even places). Difference should be multiple of 11.
Ex: $9174 \rightarrow (9+7) = 16$, $(1+4) = 5$. $16 - 5 = 11$. $11/11$.

Rule 25: 5^2 , last 2 digits should be divisible by 25. Ex: 46325, 30450

Rule 125: 5^3 , last 3 digits should be divisible by 125.

Type 1: Basic Verification

① Check if 1,848 is divisible by 3, 4 and 9.

Step 1: Check for 3 $\rightarrow 7+8+4+8 = 27/3$ (True)

Step 2: Check for 4 $\rightarrow 48/4$ (True)

Step 3: Check for 9 $\rightarrow 7+8+4+8 = 27/9$ (True)

② Test if 1659 is divisible by 7.

Soln: $1659 \rightarrow 165 - (9 \times 2) = 165 - 18 = 147$. $147/7$ (True).

③ Check if 8591 is divisible by 11.

Soln: $8591 \rightarrow \text{Sum odd: } 9+8=17$. Sum even: $5+1=6$. $17 - 6 = 11$. Divisible by 11.

④ Which number is divisible by both 9 and 11?

Soln: (a) 10,889 (b) 10,098 (c) 10,108. (Answer checked is (b) 10,098).

Type 2: Find Missing No (x, y)

⑬ Find smallest digit x such that $54x32$ is divisible by 9.

Soln: Sum = $5+4+x+3+2 = 14+x$. Smallest multiple of 9 ≥ 14 is 18. $14+x = 18 \rightarrow x = 4$.

⑭ Find all possible value of K if $734K$ is div by 6.

Soln: Div by both 3 & 2. K must be even $\{0, 2, 4, 6, 8\}$. Sum = $7+3+4+K = 14+K$. If $K=4$, $14+4 = 18$ (div by 3). $K = 4$.

⑮ The largest number which should replace x in $2365x4$ to make it divisible by 4.

⑯ What should be the value of N to make $396258N$ divisible by 8?

Soln: Last 3 digits $58N / 8 \rightarrow 584 / 8$. $N = 4$.

⑰ For what value of x is the seven digit number $46393x8$ divisible by 11?

Soln: $(4+3+8+x) - (6+9+3) = 0 \rightarrow (15+x) - 18 = 0 \rightarrow x = 3$.

⑱ If $18x3945$ is divisible by 11 where x is a digit, then x is equal to:

Soln: Sum odd = $1+x+9+5 = 15+x$. Sum even = $8+3+4 = 15$. Difference: $(15+x) - 15 = 0 \rightarrow \text{Wait, image says } 21+x \text{ and } 15$, yielding $6+x = 11 \rightarrow x = 5$.

⑲ What should replace x in $94x2357$ so that number is divisible by 11?

Soln: Sum odd = $9+x+3+7 = 19+x$. Sum even = $4+2+5 = 11$. Diff: $19+x - 11 = 8+x = 11 \rightarrow x = 3$.

Type 3: Two Unknown Digit (x, y)

- (20) If 6-digit number 342514 (written as $34x51y$ in soln) is div by 36, find x,y.

Soln: $36 = 4 \times 9$. $51y$ div by 4 $\rightarrow 512, 516$. $y \in \{2, 6\}$.

If $y = 2$: $3+4+x+5+1+2 = 15+x = 18 \rightarrow x = 3$. $(x,y) = (3,2)$.

If $y = 6$: $3+4+x+5+1+6 = 19+x = 27 \rightarrow x = 8$. $(x,y) = (8,6)$.

- (21) If an 11-digit number 545884805x6, x+y is divisible by 72, find value of $\sqrt{xy}$.

- (22) If a 9-digit number 789263x8y is div by 72, value of xy is:

Soln: $78y / 8 \rightarrow 784/8 \rightarrow y = 4$. Sum = $3+x+4$ (remainder sum).

$7+x = 9 \rightarrow x = 2$. $xy = 2 \times 4 = 8$.

- (23) If 8 digit 4432A43B is div by 9 and 5, sum of A and B is:

Soln: 4432A43B. Div by 5 $\rightarrow B = 5$ (assuming non-zero).

$A+B+2 = 9 \rightarrow A+B = 7$.

- (24) If a 10-digit number 1330x558y2 is div by 88, then value of (x+y) is:

Soln: $18 - (9+x+y) = 0$ or $11 \rightarrow 9 - (x+y) = 0 \rightarrow x+y = 9$.

Type 4: Smallest | Largest Add or Sub

- (30) What is the smallest digit to add to 4,531 to make it div by 4?

Soln: 4531. Next div by 4 is 4532. $32 - 31 = 1$. Ans = 1.

- (31) What least number must be subtract from 210, so sum is completely div by 11?

Soln: $210 / 11 = \text{Rem } (1)$. $210 - 1 = 209$ (div by 11). Ans = 1.

- (32) When 335 is added to 5A7, the result is 8B2 is div by 3. Largest possible value of A?

Soln: $335 + 5A7 = 8B2$. $A = 4$, $B = 8$.

- (33) What is the smallest digit to add to 81247 to make it div by 3?

Soln: Sum = $8+2+4+7 = 21$ (div by 3). Smallest single digit to add is 0. Ans = 0.

- (34) What is smallest number to subtract from 6583 to make it div by 5?

Soln: Last digit should be 0 or 5. Nearest lower is 6580. $6583 - 6580 = 3$. Ans = 3.

- (35) What is smallest digit to add to 5911 to make it div by 9?

Soln: Sum = $5+9+1+1 = 16$. Next mult = 18. $18 - 16 = 2$. Ans = 2.